Delta University for Sciences and Technology
Faculty of Artificial Intelligence
Artificial Intelligence Department

AI-Assisted 3D Brain Tumor Surgical Planning with Risk-Aware Trajectory Optimization

AI317 - Work-based Professional Project in Artificial Intelligence (II)

Handbook

Prepared by:

Tawfek Mohamed Tawfek	4231005
Adham Osama Alrifaie	4232010
Basmala Hashim Abdelrahman	4231371
Arwa Hisham Abdelaziz	4231468
Mariam Salah Aldin AlSayed	4231469
Malak Arfa Hussien	4231503

Supervised by:

Prof. Dr. Hanaa Salem Marie

Vice Dean of

The Faculty of Artificial Intelligence for Education and Student Affairs

Spring Semester
2025/2026

AI-Assisted 3D Brain Tumor Surgical Planning with Risk-Aware Trajectory Optimization

Abstract

This project presents an AI-based framework for brain tumor surgery planning that integrates medical image segmentation, functional brain mapping, and path optimization. The system provides an effective tool to assist in selecting safer surgical trajectories, improving preoperative decision-making. Brain tumor surgery requires highly accurate planning to maximize tumor removal while preserving critical brain functions. Traditional surgical planning methods rely heavily on manual interpretation of magnetic resonance imaging (MRI) scans, which can be time-consuming and highly dependent on the experience of the clinician. Recent advances in artificial intelligence and medical image analysis have enabled the development of automated systems capable of assisting neurosurgeons in diagnosis and surgical decision-making.

This project presents an AI-assisted framework for 3D brain tumor surgical planning with risk-aware trajectory optimization. The proposed system combines deep learning-based tumor segmentation, functional brain mapping, risk analysis, and 3D visualization into a unified pipeline. Brain MRI scans from the BraTS 2020 dataset, including T1, T1ce, T2, and FLAIR modalities, are preprocessed and analyzed using a medical image segmentation model to identify tumor regions accurately.

To enhance surgical safety, functional cortical regions such as motor, language, and visual areas are localized using a brain atlas registration approach. These regions are integrated into a 3D risk map that assigns different risk values to critical brain structures. Based on this map, multiple surgical trajectories are generated and evaluated according to path length, accumulated risk, and entry angle constraints using a risk-aware optimization strategy.

The final system provides interactive 3D visualization tools that allow users to inspect tumors, functional regions, and proposed surgical paths within an integrated web-based environment. The proposed framework aims to support neurosurgeons by improving surgical planning efficiency, reducing potential damage to eloquent brain regions, and enhancing overall clinical decision-making.

Contents

Chapter 4: System Software and Design

Chapter 5: Result and Evaluation

Chapter 6: Conclusion and Future Work

Scan to access the Project Source Code on GitHub

Link:
https://github.com/TawfekMohamed-7/AI-Assisted-3D-Brain-Tumor-Surgical-Planning-with-Risk-Aware-Trajectory-Optimization

List of Figures

List of Tables

Chapter 1
Introduction

1.1 Introduction

Brain tumors are among the most dangerous neurological diseases because they directly affect the central nervous system and critical brain functions. The human brain controls movement, sensation, speech, memory, vision, reasoning, and many other vital physiological processes. Consequently, abnormal tumor growth inside the brain may significantly affect both neurological functionality and patient quality of life.

Brain tumor diagnosis and treatment require highly accurate medical analysis and careful surgical planning. Neurosurgeons must remove as much tumor tissue as possible while minimizing damage to surrounding healthy brain regions. This task becomes increasingly difficult when tumors are located near eloquent cortical areas responsible for essential neurological functions such as movement, language, and vision [1].

Medical imaging technologies play a fundamental role in brain tumor diagnosis and surgical planning. Among these technologies, Magnetic Resonance Imaging (MRI) is considered the standard imaging modality because it provides highly detailed visualization of soft tissues without exposing patients to ionizing radiation. Multi-modal MRI scans provide complementary information about tumor structure, surrounding edema, and tissue abnormalities, enabling physicians to analyze brain tumors more accurately [2], [3].

Traditional brain tumor analysis depends heavily on manual interpretation of MRI scans by radiologists and neurosurgeons. Although experienced clinicians can perform highly accurate analysis, manual segmentation and trajectory planning remain time-consuming, subjective, and difficult to standardize. In addition, identifying tumor boundaries manually becomes increasingly challenging when tumors exhibit irregular structures or infiltrate surrounding tissues.

Recent advances in Artificial Intelligence (AI) and Deep Learning have significantly improved medical image analysis systems. Deep learning-based segmentation models have demonstrated strong performance in automatically identifying tumor regions from MRI scans with high accuracy [4], [6]. These systems help reduce manual workload while improving consistency and segmentation precision.

Modern neurosurgical planning systems increasingly integrate multiple technologies including automated segmentation, functional brain mapping, risk analysis, and trajectory optimization. Functional brain mapping identifies important cortical regions responsible for movement, speech, and visual processing, while trajectory optimization algorithms attempt to generate safer surgical paths that minimize damage to sensitive brain tissues [1], [5].

This project presents an AI-assisted framework for 3D brain tumor surgical planning with risk-aware trajectory optimization. The proposed system combines MRI preprocessing, automated tumor segmentation, functional cortex localization, risk map generation, trajectory scoring, and interactive 3D visualization within a unified medical imaging pipeline [10].

The system utilizes the BraTS 2020 dataset containing multi-modal MRI scans including T1, T1ce, T2, and FLAIR modalities[2]. Tumor segmentation is performed using deep learning techniques, while functional cortical regions are localized using atlas-based masking methods[7]. These components are integrated into a three-dimensional risk map used for evaluating multiple candidate surgical trajectories[5].

The generated trajectories are analyzed according to several criteria including path length, accumulated surgical risk, and entry angle optimization. Finally, the proposed system visualizes tumors, functional regions, and surgical paths using interactive 3D rendering tools integrated into a web-based environment designed to support neurosurgeons during preoperative planning[10].

The overall objective of this project is to improve surgical planning efficiency, reduce potential damage to eloquent brain regions, and contribute toward safer and more intelligent brain tumor surgery systems.

1.2 Brain Tumor Overview

A brain tumor is an abnormal growth of cells within the brain or surrounding tissues. Brain tumors can be classified into benign and malignant tumors depending on their growth behavior and invasiveness.

Benign tumors generally grow slowly and may not spread aggressively into surrounding tissues. However, even benign tumors can become dangerous when they compress critical brain structures or increase intracranial pressure.

Malignant tumors tend to grow rapidly and invade nearby tissues. Gliomas are among the most common malignant brain tumors and are widely studied in medical imaging research because of their complex structure and aggressive behavior[2].

Brain tumors are commonly associated with symptoms such as headaches, seizures, speech difficulties, memory loss, nausea, motor weakness, and visual disturbances. The severity of symptoms usually depends on the size and anatomical location of the tumor.

Modern medical imaging systems often divide brain tumors into multiple subregions including:

- Whole Tumor (WT)
- Tumor Core (TC)
- Enhancing Tumor (ET)
- Edema (ED)
- Necrotic / Non-Enhancing Tumor Core (NCR/NET)

Accurate identification of these regions is highly important because each tumor component may have different clinical characteristics and treatment implications[3].

1.3 Importance of MRI in Brain Tumor Analysis

Magnetic Resonance Imaging (MRI) is one of the most important imaging technologies used in brain tumor diagnosis and surgical planning. MRI provides excellent soft tissue contrast and allows physicians to observe detailed anatomical structures inside the brain.

The BraTS 2020 dataset used in this project contains four MRI modalities for each patient:

- T1
- T1ce
- T2
- FLAIR

Each modality provides unique information about brain tissues and tumor characteristics.

T1-weighted MRI provides detailed anatomical information and is useful for structural visualization. T1ce images highlight enhancing tumor regions after contrast injection. T2-weighted MRI emphasizes fluid-containing abnormalities, while FLAIR suppresses cerebrospinal fluid signals and improves edema visualization.

Combining all MRI modalities allows segmentation systems to analyze tumors more accurately because each modality contributes complementary information about tissue structure and pathological abnormalities [6], [8].

1.4 Artificial Intelligence in Medical Imaging

Artificial Intelligence has become one of the most important technologies in modern medical imaging systems. Deep learning algorithms can automatically analyze complex medical images and identify important anatomical structures with high accuracy.

In brain tumor analysis, AI models are commonly used for segmentation tasks where the system automatically identifies tumor boundaries and generates segmentation masks from MRI scans [4].

Deep learning architectures such as U-Net, SegResNet, and transformer-based segmentation models have demonstrated strong performance in brain tumor analysis challenges including the BraTS benchmark datasets[2], [4].

AI-assisted systems offer several advantages including:

- Reduced manual workload
- Faster image analysis
- Improved segmentation consistency
- Better visualization support
- Enhanced surgical planning assistance

These technologies have significantly improved the development of intelligent clinical decision-support systems for neurosurgical applications.

1.5 Problem Statement

Traditional brain tumor surgical planning methods rely heavily on manual MRI interpretation and physician expertise. This process is often time-consuming, subjective, and difficult to standardize across different clinical cases.

Although many medical imaging systems can perform tumor segmentation, fewer systems integrate segmentation, functional brain mapping, surgical risk analysis, and trajectory optimization within a single unified framework[1], [5].

Another major challenge appears when tumors are located near eloquent cortical regions responsible for movement, language, and visual processing. Selecting unsafe surgical trajectories may lead to severe postoperative neurological deficits[5].

In addition, analyzing multiple MRI modalities simultaneously increases planning complexity because physicians must mentally combine information from different image sequences.

Therefore, there is a need for an AI-assisted framework capable of:

- Automatically segmenting brain tumors
- Identifying functional brain regions
- Generating surgical risk maps
- Evaluating multiple surgical trajectories
- Providing interactive 3D visualization tools[10]

Such a system can support neurosurgeons during preoperative planning and contribute toward safer surgical interventions.

1.6 Project Objectives

The main objective of this project is to develop an AI-assisted framework for 3D brain tumor surgical planning with risk-aware trajectory optimization.

The specific objectives include:

- Building a preprocessing pipeline for multi-modal MRI scans[8]
- Segmenting brain tumor regions automatically[4]

- Identifying functional cortical regions
- Constructing a three-dimensional risk map
- Generating and evaluating multiple surgical trajectories
- Minimizing surgical risk near eloquent brain regions
- Developing interactive 3D visualization tools[10]
- Integrating all components into a unified web-based system

1.7 Project Contributions

The main contributions of this project include:

- Development of an integrated AI-assisted neurosurgical planning pipeline
- Automated segmentation of brain tumors using multi-modal MRI scans
- Functional cortex localization using atlas-based masking methods[7]
- Generation of a three-dimensional surgical risk map
- Risk-aware trajectory optimization using path scoring techniques[5]
- Interactive 3D visualization for tumors and surgical paths[10]
- Integration of the complete workflow into a web-based platform

1.8 Overall System Overview

The proposed system consists of several connected stages forming a complete surgical planning pipeline.

The process begins by loading multi-modal MRI scans from the BraTS 2020 dataset including T1, T1ce, T2, and FLAIR modalities. The MRI data are preprocessed through normalization and padding operations before segmentation[8].

After preprocessing, segmentation models identify tumor regions and generate masks representing different tumor subregions[4].

Functional brain masks are then generated for important cortical regions including motor, language, and visual cortices[7]. These masks are combined with tumor masks to construct a three-dimensional surgical risk map.

The risk map is used during trajectory planning to evaluate multiple candidate surgical paths according to path length, accumulated risk, and entry angle constraints[5].

Finally, the generated outputs are visualized using interactive 3D rendering tools integrated within a web-based interface that allows users to inspect tumors, functional regions, and candidate surgical trajectories interactively[10].

1.9 Chapter Summary

This chapter introduced the general concept of brain tumor surgical planning and discussed the importance of MRI imaging and Artificial Intelligence in modern medical imaging systems. The chapter also explained the project problem statement, objectives, contributions, and overall system overview.

The next chapter will discuss the medical and technical background of brain tumor analysis, including brain anatomy, MRI modalities, segmentation techniques, functional brain mapping, and the major clinical challenges associated with neurosurgical planning.

Chapter 2

Background and Challenges

2.1 Introduction

Brain tumor surgical planning requires the integration of multiple medical and technical fields including neuroanatomy, medical imaging, image segmentation, functional brain mapping, and trajectory optimization. Understanding the theoretical background behind these components is essential for developing intelligent systems capable of assisting neurosurgeons during preoperative planning[1].

This chapter presents the medical and technical foundations related to the proposed system. The chapter discusses brain anatomy, MRI modalities, brain tumor structures, traditional and AI-based segmentation approaches, functional cortex mapping, and the major challenges associated with neurosurgical planning systems.

In addition, the chapter highlights several limitations found in traditional surgical planning approaches and explains the motivation behind integrating Artificial Intelligence with risk-aware trajectory optimization methods[5].

2.2 Brain Anatomy Background

The human brain is an extremely complex organ responsible for controlling nearly all physiological and cognitive activities. It consists of interconnected neural structures that continuously exchange information through electrical and chemical signaling pathways.

Understanding brain anatomy is highly important in neurosurgical planning because brain tumors may appear near functional regions responsible for movement, language, sensation, or visual processing. Damage to these regions during surgery may result in severe neurological deficits.

The cerebral cortex represents the outer layer of the brain and is divided into several lobes.

2.2.1 Frontal Lobe

The frontal lobe is associated with voluntary movement, decision-making, reasoning, personality, and speech production. The motor cortex, located within the frontal lobe, controls voluntary body movement.

Tumors located near this region require highly accurate trajectory planning to avoid damaging motor pathways and surrounding cortical tissues.

2.2.2 Parietal Lobe

The parietal lobe processes sensory information including touch, pressure, pain, and spatial orientation. It contributes significantly to body awareness and sensory integration.

Surgical damage to this region may affect sensory perception and coordination abilities.

2.2.3 Temporal Lobe

The temporal lobe is involved in auditory processing, memory formation, emotional interpretation, and language comprehension.

Tumors near the temporal lobe may lead to hearing abnormalities, memory loss, or language comprehension difficulties. Consequently, preserving language-related cortical areas is essential during surgical planning.

2.2.4 Occipital Lobe

The occipital lobe is primarily responsible for visual processing. It transforms visual signals into meaningful visual perception.

Damage to visual cortex regions during surgery may result in visual impairment or partial vision loss.

2.2.5 Functional Brain Regions

Certain brain regions are classified as eloquent cortex areas because they control critical neurological functions. In this project, special focus is placed on:

- Motor Cortex
- Language Cortex
- Visual Cortex

These regions are integrated into the surgical risk map because damaging them may produce severe postoperative neurological deficits.

2.3 Brain Tumor Background

Brain tumors are abnormal masses formed by uncontrolled cellular growth inside the brain or surrounding tissues. Tumors vary significantly in shape, size, location, and aggressiveness.

Brain tumors may be classified as:

- Benign Tumors
- Malignant Tumors

Benign tumors generally grow slowly and may remain localized. Malignant tumors tend to grow rapidly and invade nearby tissues.

Gliomas are among the most common and aggressive brain tumors and are frequently studied in medical imaging research because of their infiltrative structure and complex appearance in MRI scans[2], [7].

Modern brain tumor analysis systems divide tumors into multiple subregions including:

- Whole Tumor (WT)
- Tumor Core (TC)
- Enhancing Tumor (ET)
- Edema (ED)
- Necrotic / Non-Enhancing Core (NCR/NET)

Each subregion has different radiological characteristics and clinical importance[3], [8].

2.4 MRI Modalities and Medical Imaging

Magnetic Resonance Imaging (MRI) is the primary imaging modality used in brain tumor diagnosis and analysis because it provides excellent soft tissue visualization without ionizing radiation[2].

The proposed system utilizes the BraTS 2020 dataset which contains four MRI modalities for each patient:

- T1
- T1ce
- T2
- FLAIR

Each MRI modality provides complementary information about anatomical structures and pathological abnormalities.

2.4.1 T1 MRI

T1-weighted MRI provides detailed anatomical information and structural visualization of brain tissues. It is commonly used for anatomical alignment and atlas registration.

2.4.2 T1ce MRI

T1ce images are acquired after contrast enhancement and highlight active tumor regions with abnormal vascular activity.

This modality is highly useful for identifying enhancing tumor tissues.

2.4.3 T2 MRI

T2-weighted MRI emphasizes fluid-containing tissues and abnormalities associated with edema or swelling.

Tumor-related fluid accumulation appears brighter in T2 images.

2.4.4 FLAIR MRI

FLAIR suppresses cerebrospinal fluid signals while preserving abnormal tissue visibility.

This modality improves edema detection and visualization of infiltrative tumor regions.

2.4.5 Importance of Multi-Modal MRI

Using multiple MRI modalities simultaneously improves tumor analysis because each sequence provides unique tissue information.

Combining T1, T1ce, T2, and FLAIR images allows segmentation systems to detect tumor structures more accurately than using a single modality alone [6], [8].

2.5 Traditional Brain Tumor Segmentation Methods

Before the development of deep learning systems, brain tumor segmentation mainly relied on traditional image processing techniques[4].

Common traditional methods included:

- Thresholding
- Region Growing
- Edge Detection
- Clustering Algorithms
- Atlas-Based Segmentation

Thresholding methods attempted to separate tumor tissues based on intensity values. However, MRI intensity variations often reduced segmentation accuracy.

Region growing approaches expanded segmented areas from selected seed points, but these methods were sensitive to noise and intensity inhomogeneity.

Clustering techniques such as K-Means grouped pixels according to intensity similarities. Although useful in some cases, clustering methods struggled with complex tumor structures and overlapping tissue appearances.

Atlas-based segmentation methods used predefined anatomical templates for tissue localization. However, anatomical variations between patients often reduced generalization performance.

Traditional segmentation methods generally suffered from several limitations including:

- Sensitivity to noise
- Poor generalization
- Difficulty handling irregular tumors
- Dependence on handcrafted features
- Limited accuracy near tumor boundaries

These limitations motivated the adoption of deep learning-based segmentation techniques.

2.6 Deep Learning in Brain Tumor Segmentation

Deep learning has significantly improved medical image analysis and segmentation performance during recent years[4], [6].

Convolutional Neural Networks (CNNs) automatically learn hierarchical image features directly from training data without requiring handcrafted feature engineering.

Several deep learning architectures have demonstrated strong performance in brain tumor segmentation tasks including:

- U-Net
- SegResNet
- Transformer-Based Models

These architectures achieved high segmentation accuracy on benchmark datasets such as BraTS [2], [7].

2.6.1 U-Net Architecture

U-Net is one of the most widely used architectures in medical image segmentation because of its encoder-decoder structure and skip connections.

The encoder extracts image features while the decoder reconstructs segmentation masks with detailed spatial information.

U-Net performs well on medical datasets because it can capture both local and global image features effectively.

2.6.2 SegResNet Architecture

SegResNet combines residual learning with segmentation networks to improve feature extraction and training stability.

Residual connections help preserve gradient flow during training and improve segmentation performance in deep neural networks.

SegResNet has demonstrated strong performance in medical imaging tasks involving multi-modal MRI analysis[8].

2.6.3 Transformer-Based Models

Transformer-based architectures have recently gained attention in medical imaging because they can model long-range spatial relationships within images.

These models improve contextual understanding and capture global image dependencies more effectively than conventional CNNs in certain tasks[6], [9].

2.7 Functional Brain Mapping

Functional brain mapping is the process of identifying cortical regions responsible for essential neurological functions such as movement, language, and vision.

In this project, functional mapping is performed using atlas-based masking techniques. Functional masks are generated for:

- Motor Cortex
- Language Cortex

- Visual Cortex

These masks are aligned with patient MRI scans and integrated into the risk map generation stage.

Functional mapping is highly important because tumors may develop near eloquent cortical regions. Surgical damage to these regions may produce severe postoperative deficits[1], [5].

2.8 Surgical Planning Challenges

Brain tumor surgical planning is associated with several major challenges.

One major challenge involves selecting safe surgical trajectories that minimize damage to surrounding healthy tissues and functional brain regions.

Another challenge arises from the irregular structure of brain tumors. Tumors may infiltrate nearby tissues, making boundary identification difficult even for experienced clinicians.

In addition, multiple MRI modalities must often be analyzed simultaneously, increasing interpretation complexity.

Functional cortical regions create additional surgical constraints because trajectories passing through eloquent cortex areas may significantly increase surgical risk.

Manual trajectory selection also depends heavily on physician expertise and subjective interpretation.

These challenges motivate the development of AI-assisted systems capable of integrating segmentation, functional mapping, risk analysis, and trajectory optimization into a unified framework [1], [5].

2.9 Risk-Aware Surgical Planning

Risk-aware surgical planning aims to generate safer trajectories by minimizing surgical interaction with critical brain structures.

In the proposed system, different brain regions are assigned different risk values according to their surgical importance.

Functional cortical regions receive higher risk values because damaging these regions may produce severe neurological consequences.

The tumor itself receives the lowest risk value because it represents the surgical target that must be removed.

A three-dimensional risk map is generated by combining tumor masks, functional masks, and surrounding brain structures.

This risk map is later used during trajectory scoring and optimization[5].

2.10 Existing Limitations and Research Motivation

Although many brain tumor segmentation systems have achieved high accuracy, fewer systems integrate segmentation with functional brain mapping and trajectory optimization within a complete surgical planning framework.

Many existing approaches focus only on segmentation performance without considering surgical safety or trajectory analysis.

In addition, several systems lack interactive visualization tools that allow physicians to inspect tumors and surgical paths in three-dimensional space.

These limitations motivated the development of the proposed AI-assisted framework which combines:

- Tumor segmentation
- Functional cortex localization
- Risk-aware trajectory optimization
- Interactive 3D visualization
- Web-based integration

within a unified medical imaging pipeline[1], [4], [5].

2.11 Chapter Summary

This chapter discussed the medical and technical background related to brain tumor surgical planning systems. Brain anatomy, MRI modalities, tumor structures, traditional segmentation approaches, deep learning methods, and functional brain mapping were presented.

The chapter also discussed major clinical and technical challenges associated with neurosurgical planning and highlighted the motivation behind developing AI-assisted risk-aware surgical planning systems.

The next chapter will present the proposed framework and methodology including dataset preparation, preprocessing, segmentation, functional mapping, risk map generation, trajectory planning, and visualization techniques.

Chapter 3
Framework and Methodology

3.1 Introduction

This chapter presents the proposed framework and methodology used for AI-assisted 3D brain tumor surgical planning with risk-aware trajectory optimization. The framework integrates multiple medical imaging and computational stages into a unified pipeline designed to support neurosurgeons during preoperative planning[1], [5].

The proposed system combines MRI preprocessing, deep learning-based tumor segmentation, functional cortex localization, atlas registration, surgical risk analysis, trajectory generation, path scoring, and interactive three-dimensional visualization[4], [5]. The complete workflow of the system is illustrated in Figure 3.1.

The primary objective of the framework is not only to segment brain tumors accurately, but also to generate safer surgical trajectories by considering the spatial relationship between tumors and critical brain regions. Unlike conventional segmentation-only systems, the proposed framework integrates anatomical risk information into the trajectory planning process[5].

The workflow starts with loading multi-modal MRI scans and atlas data. The MRI volumes then pass through several preprocessing operations to standardize orientation, voxel spacing, and intensity distribution. After preprocessing, the SegResNet segmentation model identifies tumor regions and generates tumor masks. Functional cortical regions and white matter tracts are then aligned with patient MRI scans using atlas registration techniques[8].

The resulting anatomical information is combined into a voxel-wise surgical risk map used during candidate trajectory generation and path evaluation. Finally, the generated trajectories and anatomical structures are visualized within an interactive three-dimensional environment designed to support surgical interpretation and planning[1], [5].

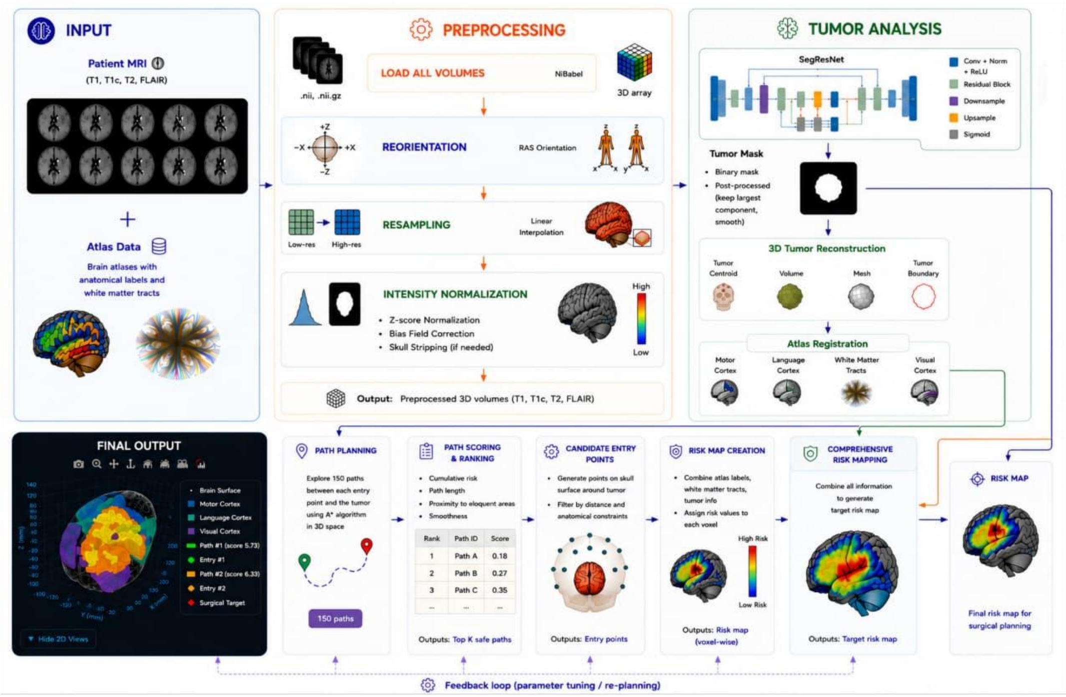

Figure 3.1: Proposed AI-Assisted Surgical Planning Framework

As shown in Figure 3.1, the framework consists of several interconnected stages:

1. Input Data Acquisition
2. MRI Preprocessing
3. Tumor Segmentation and Analysis
4. Atlas Registration
5. Functional Cortex Mapping

6. Risk Map Creation
7. Candidate Entry Point Generation
8. Surgical Path Planning
9. Path Scoring and Ranking
10. Interactive 3D Visualization

Each stage contributes to improving surgical safety and generating more reliable trajectory recommendations.

3.2 Input Data

The proposed framework relies on two major categories of input data:

- Multi-modal MRI scans
- Brain atlas data

The integration of both anatomical imaging and atlas-based functional information allows the framework to perform both tumor analysis and risk-aware trajectory planning.

3.2.1 MRI Data

The MRI data used in this project are obtained from the BraTS 2020 dataset. The dataset contains multi-modal MRI scans for brain tumor patients and is widely used in medical image segmentation research [2], [6].

Each patient contains four MRI modalities:

Table 3.1 MRI Modalities Used in the Framework

MRI Modality	Function
T1	structural anatomical information
T1ce	Highlights enhancing tumor regions
T2	Emphasizes fluid-containing abnormalities
FLAIR	Improves edema visualization

Each modality provides different anatomical and pathological information about the brain and tumor structures.

The MRI scans are stored using medical imaging formats including:

- .nii
- .nii.gz

The system loads all MRI modalities as three-dimensional volumes using medical imaging libraries capable of handling volumetric neuroimaging data.

The use of multi-modal MRI is essential because no single modality contains sufficient information for accurate tumor analysis. Instead, different modalities provide complementary information that improves segmentation quality and anatomical understanding.

For example:

- T1 images provide structural anatomical information.
- T1ce images highlight enhancing tumor regions.
- T2 images emphasize fluid-containing abnormalities.
- FLAIR images improve edema visualization.

Combining all modalities improves both segmentation performance and surgical planning reliability[6], [8].

3.2.2 Atlas Data

In addition to MRI scans, the framework uses atlas data containing anatomical labels and white matter tract information.

The atlas includes labels corresponding to important functional regions including:

- Motor Cortex
- Language Cortex
- Visual Cortex
- White Matter Tracts

These structures are considered surgically sensitive because damage to these regions may produce severe postoperative neurological deficits.

The atlas data are integrated into the risk analysis stage to allow the framework to identify anatomically safe and unsafe regions during trajectory generation.

Atlas-based functional mapping improves surgical planning by providing additional anatomical context beyond tumor segmentation alone[1], [5].

3.3 MRI Preprocessing

MRI preprocessing is one of the most important stages of the proposed framework because raw MRI scans often contain variations in orientation, voxel spacing, and intensity distribution.

Without preprocessing, these inconsistencies may negatively affect segmentation accuracy and atlas registration quality[6].

The preprocessing pipeline standardizes MRI volumes before segmentation and analysis. As illustrated in Figure 3.1, the preprocessing stage consists of:

Table 3.2 MRI Preprocessing Stages

Stage	Purpose
Volume Loading	Load MRI modalities into the system
Reorientation	Standardize anatomical orientation
Resampling	Unify voxel spacing
Intensity Normalization	Reduce intensity variations

The output of this stage is a set of standardized three-dimensional MRI volumes ready for tumor analysis.

3.3.1 Load All Volumes

The first preprocessing step involves loading all MRI modalities corresponding to each patient.

Each MRI modality is converted into a three-dimensional array representation suitable for voxel-based analysis.

The modalities are loaded simultaneously to preserve spatial alignment between MRI sequences. Maintaining this alignment is essential because segmentation models analyze information across multiple modalities at corresponding spatial locations.

The loaded MRI volumes represent the complete anatomical input for subsequent processing stages.

3.3.2 Reorientation

MRI scans obtained from different imaging systems may use different anatomical orientations.

Differences in orientation can lead to spatial inconsistencies during segmentation and atlas registration. To solve this issue, the framework performs reorientation into a standardized RAS orientation format.

RAS orientation ensures that all MRI scans follow a consistent anatomical coordinate system:

- Right
- Anterior
- Superior

Standardizing orientation improves spatial consistency across all patient data and simplifies downstream processing operations.

Reorientation is particularly important when integrating atlas masks with patient MRI volumes because anatomical labels must align correctly with brain structures.

3.3.3 Resampling

MRI scans may contain different voxel resolutions depending on acquisition settings and scanner configurations.

Different voxel sizes may negatively affect segmentation consistency and spatial alignment between modalities. Therefore, the framework applies resampling operations to standardize voxel spacing.

Linear interpolation is used during resampling to preserve anatomical continuity while adjusting spatial resolution.

Resampling improves:

- Spatial consistency
- Atlas alignment accuracy
- Segmentation stability
- Three-dimensional reconstruction quality

After resampling, all MRI modalities share consistent voxel dimensions suitable for voxel-wise analysis.

3.3.4 Intensity Normalization

MRI intensity distributions often vary significantly between patients and imaging systems.

These variations may cause deep learning models to focus on scanner-specific intensity differences rather than anatomical structures.

To reduce intensity inconsistencies, the framework applies several normalization techniques including:

- Z-score normalization
- Bias field correction
- Skull stripping (if needed)

Z-score normalization standardizes voxel intensities by centering the intensity distribution around the mean intensity value.

Bias field correction reduces low-frequency intensity variations caused by scanner artifacts.

Skull stripping may also be applied when necessary to remove non-brain tissues surrounding the brain volume.

Normalization improves segmentation robustness and allows the segmentation model to analyze MRI scans more consistently across different patient cases[4], [6].

At the end of preprocessing, the framework produces standardized MRI volumes ready for segmentation and risk analysis.

3.4 Tumor Analysis

The tumor analysis stage is responsible for detecting tumor regions and reconstructing tumor structures in three-dimensional space.

As illustrated in Figure 3.1, the tumor analysis stage includes:

- Segmentation using SegResNet
- Tumor mask generation
- Post-processing
- 3D tumor reconstruction

This stage transforms preprocessed MRI volumes into anatomically meaningful tumor representations used during surgical planning[4].

3.4.1 Segmentation Using SegResNet

The proposed framework uses the SegResNet architecture for brain tumor segmentation[8].

SegResNet combines:

- Convolutional layers
- Normalization layers
- Residual blocks
- Downsampling operations
- Upsampling operations

The model processes multi-modal MRI volumes and generates tumor segmentation masks representing tumor regions.

Residual connections improve gradient propagation and feature extraction during segmentation.

The segmentation model identifies abnormal tumor tissues from surrounding healthy brain structures and generates binary tumor masks used during subsequent planning stages.

Segmentation accuracy is highly important because trajectory planning depends directly on tumor localization quality.

Incorrect segmentation may lead to inaccurate risk analysis and unsafe trajectory generation[5].

3.4.2 Tumor Mask Generation

The output of the segmentation model is converted into binary tumor masks representing detected tumor structures.

Post-processing operations are then applied to improve segmentation quality.

These operations include:

- Keeping the largest connected component
- Smoothing operations

Keeping the largest connected component helps remove isolated noisy regions that do not belong to the actual tumor.

Smoothing operations improve tumor boundary continuity and reduce irregular segmentation artifacts.

The resulting tumor masks provide cleaner anatomical structures suitable for three-dimensional reconstruction and trajectory analysis[4].

3.4.3 3D Tumor Reconstruction

After segmentation, the framework reconstructs tumor structures in three-dimensional space.

This stage extracts important tumor properties including:

- Tumor centroid
- Tumor volume
- Tumor mesh
- Tumor boundary

The tumor centroid represents the approximate geometric center of the tumor and is later used as the target point during surgical trajectory generation.

Tumor meshes provide surface representations used during three-dimensional visualization.

Three-dimensional reconstruction improves anatomical understanding and allows physicians to inspect tumor geometry interactively[5].

3.5 Atlas Registration and Functional Mapping

Atlas registration aligns functional brain masks with patient MRI volumes.

The framework registers atlas regions corresponding to:

Table 3.3 Functional Structures Used for Risk Analysis

Structure	Importance
Motor Cortex	Related to movement control
Language Cortex	Related to speech and language
Visual Cortex	Related to vision
White Matter Tracts	Important neural pathways

The registration process spatially aligns atlas masks with the patient anatomy.
This alignment allows the framework to identify critical functional structures relative to the tumor location.
Functional mapping is highly important because tumors may appear near eloquent cortical regions responsible for essential neurological functions.
Damage to these regions during surgery may produce severe complications including:

- Motor impairment
- Speech deficits
- Visual dysfunction

The registered functional masks are later integrated into the surgical risk map[1], [5].

3.6 Risk Map Creation

The proposed framework generates a voxel-wise three-dimensional surgical risk map.
The purpose of the risk map is to represent the spatial distribution of surgical risk across the brain volume.
As illustrated in Figure 3.1, the risk map combines:

Table 3.4 Components of the Surgical Risk Map

Component	Specifications

Tumor Masks	Define tumor location
Functional Masks	Identify eloquent brain regions
White Matter Tracts	Represent critical neural pathways
Atlas Labels	Provide anatomical reference

Each voxel receives a risk value according to anatomical importance.
Critical functional regions receive higher risk values because damage to these structures may lead to severe neurological consequences.
The tumor itself receives the lowest risk value because it represents the intended surgical target.
The generated risk map provides an anatomical safety representation used during trajectory evaluation and optimization [5].

3.7 Comprehensive Risk Mapping

After generating voxel-level risk values, the framework combines all available anatomical information into a comprehensive target risk map.
The comprehensive map integrates:

- Functional cortex masks
- White matter tracts
- Tumor structures
- Anatomical labels

This stage produces the final risk representation used during path planning.
The comprehensive risk map allows the framework to distinguish between:

- Safe regions
- Moderate-risk regions
- High-risk functional areas

The final map serves as the core component for risk-aware surgical trajectory optimization [5].

3.8 Candidate Entry Point Generation

The framework generates multiple candidate surgical entry points on the skull surface surrounding the tumor.

The objective of this stage is to identify realistic starting locations for surgical trajectories.

Generated entry points are filtered according to:

- Distance constraints
- Anatomical constraints

Entry points violating safety conditions or unrealistic geometric constraints are removed.

The remaining candidate points serve as valid trajectory starting locations during path planning.

Generating multiple entry points increases the probability of finding safer trajectories that minimize interaction with critical functional structures[1], [5].

3.9 Surgical Path Planning

The path planning stage generates possible surgical trajectories between skull entry points and the tumor target.

The framework explores multiple candidate paths within three-dimensional space.

Each path connects:

- A candidate entry point
- The tumor centroid

Trajectory generation considers the spatial relationship between the tumor and surrounding anatomical structures.

Instead of selecting trajectories solely based on shortest distance, the framework integrates anatomical risk information into trajectory generation.

This allows the system to prioritize safer paths that avoid functional cortical regions and white matter tracts.

The generated trajectories are later evaluated using risk-aware scoring methods[5].

3.10 Path Scoring and Ranking

After generating candidate trajectories, the framework evaluates each path according to several surgical criteria.

The scoring process considers:

Table 3.5 Path Evaluation Criteria

Criterion	Description
Cumulative Risk	Total risk along trajectory
Path Length	Distance from entry point to tumor
Proximity to Eloquent Regions	Distance from critical brain areas
Path Smoothness	Practicality of surgical access
Printer	Windows 2000/Windows XP/Windows Vista/Windows 7 Windows 8 compatible printers
Operating System	Windows 2000 / Windows XP Windows Vista / Windows 7 Windows 8

Paths intersecting high-risk regions receive higher cumulative risk scores.

Longer trajectories may increase surgical complexity and tissue damage.

Trajectory smoothness is also important because smoother paths are generally more practical for surgical access.

After evaluation, all trajectories are ranked according to overall safety and efficiency scores.

The framework outputs the safest candidate trajectories for visualization and surgical analysis.

3.11 Interactive 3D Visualization

The final stage of the framework visualizes all generated outputs using an interactive three-dimensional environment.

The visualization system displays:

- Brain surface
- Tumor structures
- Functional cortical regions
- Candidate entry points
- Surgical trajectories
- Risk maps

The interactive interface allows users to inspect tumors and trajectories from different viewing angles.

Visualization improves anatomical understanding and supports surgical interpretation.

The final visualization environment provides an integrated representation of tumor anatomy, functional regions, and optimized surgical trajectories[5].

3.12 Final Output

The final output of the proposed framework consists of:

- Segmented tumor masks
- Three-dimensional tumor reconstruction
- Functional cortical localization

- Surgical risk maps
- Ranked candidate trajectories
- Interactive 3D visualization results

These outputs collectively support AI-assisted neurosurgical planning and provide safer trajectory recommendations based on anatomical risk analysis[1], [5].

3.13 Chapter Summary

This chapter presented the proposed framework and methodology used for AI-assisted brain tumor surgical planning with risk-aware trajectory optimization.

The chapter discussed the complete processing pipeline including MRI preprocessing, SegResNet-based tumor segmentation, atlas registration, functional cortex mapping, risk map

generation, trajectory planning, path scoring, and interactive three-dimensional visualization.

The integration of tumor analysis, functional mapping, and risk-aware optimization allows the framework to support safer and more intelligent neurosurgical planning[1], [5].

Chapter 4

System Software and Design

4.1 Introduction

This chapter presents the software architecture and system design of the proposed AI-assisted brain tumor surgical planning system. The proposed system integrates a mobile application interface, authentication services, backend processing modules, and AI-assisted medical imaging components within a unified architecture[1], [5].

The system is designed to provide an interactive and user-friendly environment that allows users to navigate through different sections of the application while interacting with the brain tumor surgical planning framework. The architecture combines Flutter-based mobile application development, Firebase authentication services, and FastAPI backend communication[5].

The frontend application is responsible for user interaction, navigation, and interface management, while the backend handles communication with the AI processing pipeline responsible for segmentation, risk analysis, and trajectory planning[1], [5].

The proposed architecture aims to provide:

- User-friendly interaction
- Organized navigation
- Secure authentication
- Efficient backend communication
- Integration with AI-assisted processing modules
- Support for visualization and surgical planning workflows

This chapter discusses the overall architecture, frontend implementation, authentication system, backend integration, data flow, and user interaction workflow.

4.2 Overall System Architecture

The proposed system consists of three major components:

Table 4.1 Main System Components

Component	Function
Flutter	User interface and navigation
Firebase	Authentication and user management
FastAPI	Backend communication and processing

These components work together to provide an integrated environment for AI-assisted surgical planning[5].

The Flutter mobile application serves as the frontend layer responsible for user interaction and interface navigation. Firebase is used for authentication management including login and registration operations. The FastAPI backend manages communication with the AI processing modules responsible for tumor analysis and surgical planning[1], [5].

The general workflow of the system can be summarized as follows:

1. The user opens the mobile application.
2. Authentication is performed using Firebase.
3. The user navigates through the application sections.
4. Requests are sent from the frontend to the FastAPI backend.
5. The backend communicates with the AI processing pipeline.
6. Generated outputs are returned to the application interface.

The architecture separates frontend interaction from backend processing, allowing the system to remain modular and scalable[5].

4.3 Flutter Mobile Application

The frontend of the proposed system is developed using Flutter.

Flutter provides a cross-platform development environment capable of building interactive and responsive mobile applications. The framework supports organized interface design, navigation management, and integration with backend services[5].

The mobile application acts as the primary user interaction layer of the proposed system. Users interact with the application through multiple screens and navigation components designed to simplify access to different sections of the project.

The application contains several screens including:

- Splash Screen
- Login Screen
- Register Screen
- Main Navigation Screens

The frontend structure is organized to provide smooth navigation and improve user experience[5].

4.4 Splash Screen Design

The Splash Screen represents the initial interface displayed when the application starts.

This screen serves several purposes including:

- Displaying the application identity
- Initializing application resources
- Preparing navigation flow
- Improving user experience during startup

The splash screen acts as the entry point to the application before transitioning to the authentication screens.

Using a dedicated splash screen improves application organization and creates a more professional user experience.

4.5 Authentication System

The proposed system uses Firebase Authentication for user login and registration management.

Firebase provides ready-to-use authentication APIs that simplify secure user management within the application.

The authentication module includes:

- Login functionality
- Registration functionality
- User session management

The authentication process ensures that users can securely access the application before interacting with the system features.

4.5.1 Login Screen

The login screen allows registered users to access the application using their authentication credentials.

The login interface communicates directly with Firebase authentication services to validate user information.

After successful authentication, users are redirected to the main application interface.

The login system improves application security and provides controlled access to the system.

4.5.2 Register Screen

The register screen allows new users to create accounts within the application.

The registration process is connected directly to Firebase authentication services.
User information is validated before account creation to ensure correct authentication flow.
After successful registration, users can access the application using their newly created credentials.
The registration system improves accessibility and allows multiple users to interact with the application.

4.6 Navigation System

The proposed mobile application uses a Bottom Navigation Bar for screen navigation.
The navigation bar provides organized movement between the primary application sections and improves overall usability.
Using bottom navigation simplifies user interaction because users can quickly switch between different sections without leaving the main interface.
The navigation structure is designed to maintain smooth interaction flow throughout the application.
The primary sections accessible through the navigation system include:

- Home
- Overview
- Get Started
- About Us
- Contact

The navigation design focuses on simplicity and accessibility while maintaining organized screen transitions.

4.7 Frontend User Interface Design

The frontend interface is divided into multiple sections designed to organize project information and user interaction.

Each section provides specific functionality related to the proposed system.

4.7.1 Home Section

The Home section acts as the primary landing interface after authentication.

This section provides a general introduction to the project and presents the main purpose of the proposed AI-assisted surgical planning system.

The Home interface improves user orientation and provides an overview of the application environment.

4.7.2 Overview Section

The Overview section presents general information about the project and its workflow.

This section explains the overall concept of AI-assisted brain tumor surgical planning and introduces the main processing stages of the framework.

The overview interface helps users understand the functionality and objectives of the proposed system before interacting with additional features.

4.7.3 Get Started Section

The Get Started section represents the primary interaction area related to the project workflow.

This section is designed to guide users toward interacting with the proposed surgical planning framework.

The Get Started interface acts as the connection point between the frontend environment and the backend processing system.

This section improves usability by organizing project interaction within a dedicated interface environment.

4.7.4 About Us Section

The About Us section provides information about the project team and the development background of the proposed system.

This section helps present the contributors involved in the development of the project.

Including an About Us section improves application organization and provides contextual information about the system creators.

4.7.5 Contact Section

The Contact section provides communication information related to the project.

This section allows users to identify communication channels associated with the application and project team.

The Contact interface contributes to improving accessibility and user engagement.

4.8 FastAPI Backend Architecture

The backend of the proposed system is implemented using FastAPI.

FastAPI acts as the communication layer between the mobile application and the AI-assisted processing modules[5].

The backend architecture is responsible for:

- Receiving frontend requests
- Managing API communication
- Triggering processing operations
- Returning generated outputs

Using FastAPI improves backend performance and supports organized API structure.

The backend architecture separates interface logic from computational processing, improving modularity and maintainability[5].

4.9 Backend and AI Pipeline Integration

The FastAPI backend communicates with the AI processing pipeline responsible for medical imaging analysis and surgical planning operations[1], [5].

The backend acts as an intermediary layer connecting frontend interaction with backend computational modules.

The AI pipeline includes several processing stages including:

- MRI preprocessing
- Tumor segmentation
- Functional mapping
- Risk map generation
- Path planning
- Visualization generation

The backend coordinates communication between these modules and organizes data transfer across the processing workflow.

This integration allows the mobile application to interact with the AI-assisted planning system through a unified backend environment[5].

4.10 Frontend and Backend Communication

Communication between the Flutter frontend and FastAPI backend occurs through API requests and responses.

The frontend sends requests to the backend whenever processing operations or system interactions are required.

The backend processes incoming requests and returns generated outputs to the mobile application.

This communication structure improves modularity because the frontend and backend remain logically separated.

Separating frontend and backend components provides several advantages including:

- Easier maintenance
- Better scalability
- Independent component development
- Improved system organization

The API-based communication model also supports future system extensions and additional functionality integration[5].

4.11 Application Workflow

The proposed system follows an organized interaction workflow beginning from application startup until user navigation through the project sections.

The workflow can be summarized as follows:

1. The user launches the mobile application.
2. The splash screen is displayed.
3. The user navigates to login or registration.
4. Firebase authentication validates user credentials.
5. The user accesses the main application interface.
6. Navigation occurs through the Bottom Navigation Bar.
7. The user explores different application sections.
8. Frontend requests are sent to the FastAPI backend when required.
9. Backend processing operations are executed.
10. Generated outputs are returned to the application interface.

This workflow provides organized interaction between users and the proposed surgical planning system.

4.12 System Design Considerations

Several design considerations were taken into account during development of the proposed system.

One important consideration was usability. The frontend interface was organized using multiple dedicated sections and a bottom navigation structure to simplify user interaction.

Another consideration involved modularity. The separation between frontend, authentication services, and backend processing improves system maintainability and future extensibility.

Security was also considered through the use of Firebase authentication services for login and registration management.

In addition, backend integration using FastAPI improves communication organization between the frontend environment and AI-assisted processing modules.

These design decisions collectively contribute to building a more organized and scalable application architecture[5].

4.13 System Advantages

The proposed software architecture provides several advantages including:

- Organized mobile application interface
- Cross-platform frontend development using Flutter
- Secure authentication using Firebase
- Structured backend communication using FastAPI
- Modular system architecture
- Scalable application structure
- Simplified navigation workflow
- Integration with AI-assisted medical imaging pipeline

The combination of frontend usability and backend modularity improves the overall efficiency and usability of the proposed system[5].

4.14 Chapter Summary

This chapter presented the software architecture and system design of the proposed AI-assisted brain tumor surgical planning system.

The chapter discussed the Flutter mobile application, Firebase authentication integration, FastAPI backend architecture, frontend navigation system, application sections, backend communication workflow, and overall system interaction design.

The integration of frontend interaction, authentication services, and backend processing provides a modular and organized environment capable of supporting AI-assisted surgical planning workflows[1], [5].

Chapter 5

Results and Evaluation

5.1 Introduction

This chapter presents the experimental results and evaluation of the proposed AI-assisted brain tumor surgical planning framework. The evaluation process focuses primarily on the performance of the tumor segmentation stage, model training behavior, qualitative prediction quality, and overall system functionality.

The proposed framework was evaluated using MRI data from the BraTS 2020 dataset[6], [2]. The experiments focused on assessing the ability of the segmentation model to identify tumor regions accurately while maintaining stable training performance throughout the optimization process.

Several visualization outputs and evaluation plots were generated during experimentation including:

- MRI slice visualization
- Label overlay visualization
- Training loss curves
- Validation Dice score curves
- Ground truth and prediction comparisons

The purpose of this chapter is to analyze the effectiveness of the proposed framework and demonstrate the capability of the segmentation model to identify tumor structures from multi-modal MRI data.

5.2 Dataset Visualization and Label Representation

Before training the segmentation model, MRI scans and tumor labels were visually inspected to verify dataset quality and label consistency.
Figure 5.1 illustrates an example of a T1CE MRI slice together with the corresponding tumor label overlay[2], [8].

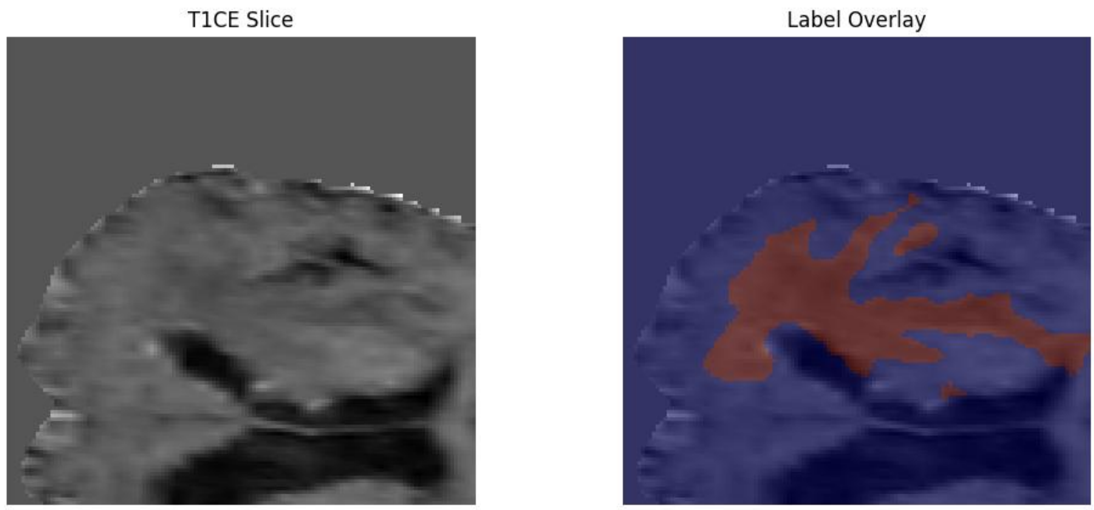

Figure 5.1: T1CE MRI Slice and Tumor Label Overlay

As shown in Figure 5.1, the left image represents a T1CE MRI slice extracted from the patient volume, while the right image shows the corresponding tumor label overlay.
The overlay visualization highlights tumor regions using colored label masks placed directly over the MRI image. This visualization process is important because it allows verification of spatial alignment between MRI scans and tumor annotations before training the segmentation model.
The tumor regions appear concentrated within localized brain areas, while surrounding healthy tissues remain unlabeled. The visualization also demonstrates the complexity of tumor boundaries and the irregular anatomical structure of tumor regions [3], [7].

Visual inspection of MRI slices and labels plays an important role during preprocessing and dataset validation because inaccurate labels or misaligned masks may negatively affect segmentation performance.
The overlay representation additionally helps verify that the preprocessing pipeline preserves anatomical consistency between MRI modalities and segmentation masks.

5.3 Model Training Performance

The segmentation model was trained using the preprocessed MRI volumes obtained from the BraTS 2020 dataset.

During training, the model parameters were optimized iteratively while monitoring training performance across epochs. One of the primary indicators used during training evaluation was the training loss curve.

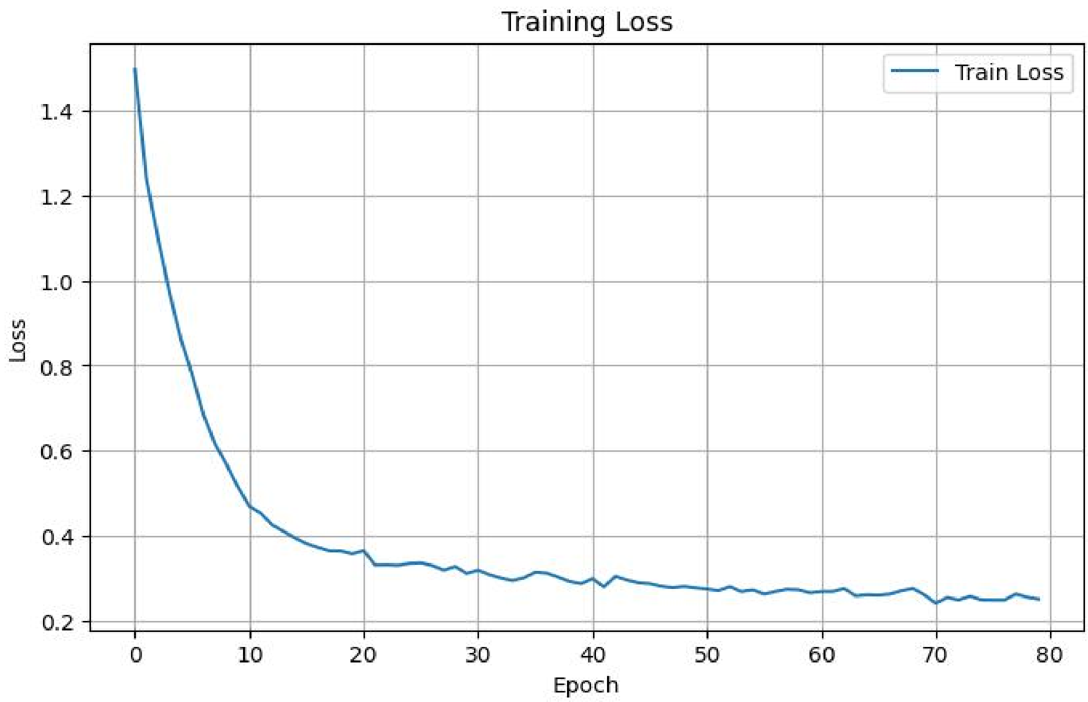

Figure 5.2: Training Loss During Model Training

As illustrated in Figure 5.2, the training loss was initially high during the early epochs of training. This behavior is expected because the segmentation model begins training with randomly initialized parameters and limited understanding of tumor structures.

As training progressed, the loss value decreased gradually, indicating that the model was learning meaningful anatomical and pathological features from the MRI data[4], [6].

A rapid reduction in loss occurred during the initial training epochs, followed by a slower and more stable decrease during later epochs. This behavior suggests that the model progressively improved segmentation quality while approaching convergence.

The training curve eventually became relatively stable during later epochs, which indicates improved optimization stability and reduced parameter fluctuations.

Stable training behavior is important because unstable optimization may produce inconsistent segmentation results and poor generalization performance.

The gradual reduction in training loss demonstrates that the model successfully learned tumor-related features from the MRI volumes and improved prediction quality over time.

5.4 Segmentation Evaluation Using Dice Score

The segmentation performance of the proposed framework was evaluated using the Dice similarity metric[4], [8].

Dice score is commonly used in medical image segmentation tasks because it measures the overlap between predicted segmentation masks and ground truth labels[4].

Higher Dice scores indicate stronger agreement between predictions and reference tumor masks.

Figure 5.3 illustrates the validation mean Dice score measured during training epochs.

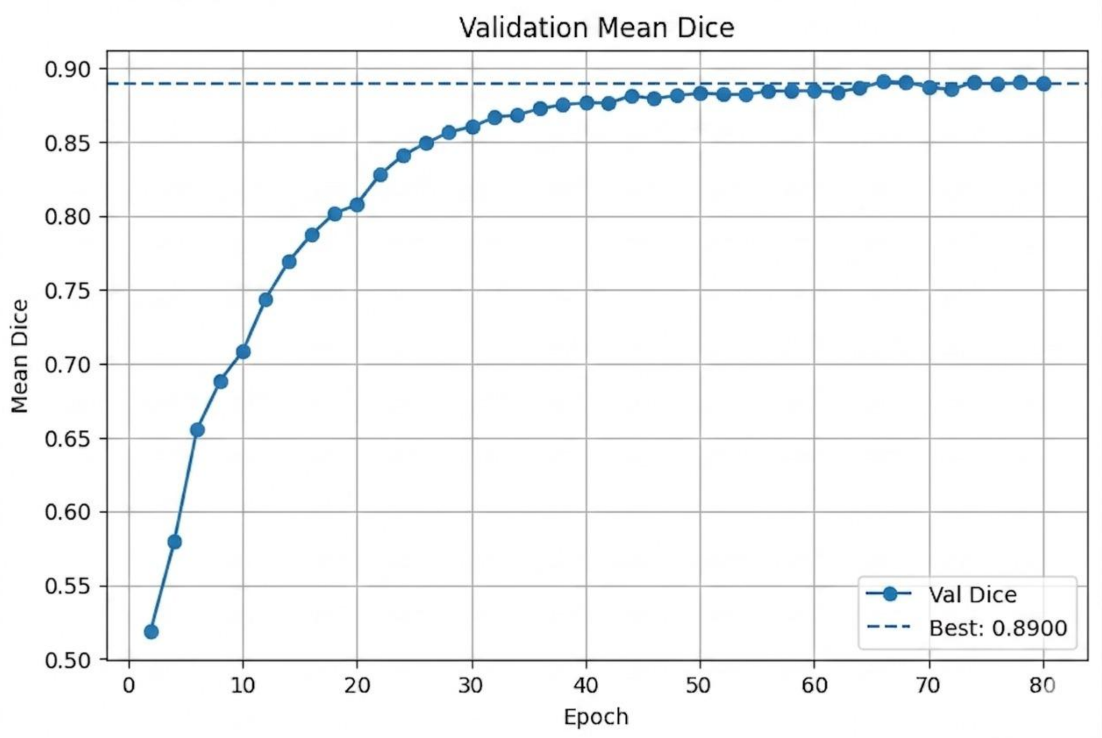

Figure 5.3: Validation Mean Dice Score

As shown in Figure 5.3, the validation Dice score improved progressively during training.
At the beginning of training, the Dice score was relatively low because the model had not yet learned sufficient tumor features. However, after several epochs, the Dice score increased steadily, indicating continuous improvement in segmentation accuracy.

The validation Dice score eventually approached approximately 0.89, demonstrating that the segmentation model achieved relatively strong overlap between predicted tumor regions and ground truth annotations.
The gradual increase in Dice score also indicates improved model generalization across validation samples.

The stabilization of the validation curve during later epochs suggests that the model reached a relatively stable learning stage without significant performance oscillations.

The close relationship between decreasing training loss and increasing Dice score confirms that the optimization process successfully improved segmentation quality throughout training.

5.5 Qualitative Segmentation Results

In addition to quantitative evaluation metrics, qualitative visual comparisons were performed to evaluate segmentation quality.

Figure 5.4 presents a comparison between:

- Original MRI slice
- Ground truth segmentation
- Predicted segmentation

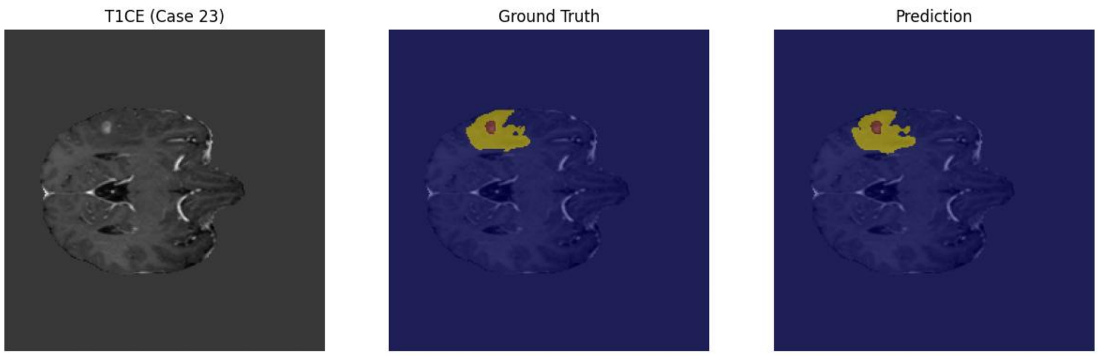

Figure 5.4: Ground Truth and Predicted Tumor Segmentation

As illustrated in Figure 5.4, the left image represents the original T1CE MRI slice, while the middle image represents the ground truth segmentation mask. The right image shows the predicted tumor segmentation generated by the model.

The prediction output demonstrates strong visual similarity to the reference ground truth mask. The predicted segmentation successfully identified the primary tumor region and preserved the overall anatomical structure of the tumor boundaries.

The segmentation model was capable of localizing the tumor region while maintaining reasonable boundary consistency relative to the ground truth labels.

Although small variations may exist between predicted and reference masks, the overall overlap remains visually consistent, indicating effective segmentation performance[6], [8].

Qualitative visualization is particularly important in medical imaging because visual inspection helps evaluate anatomical realism in addition to numerical performance metrics.

The comparison additionally demonstrates the capability of the segmentation framework to detect tumor structures from MRI scans with relatively accurate spatial localization.

5.6 Discussion of Model Performance

The experimental results demonstrate that the proposed segmentation framework achieved stable training behavior and effective tumor localization performance.

Several observations can be made from the experimental results:

- The training loss decreased consistently throughout training.
- The validation Dice score improved progressively across epochs.
- The segmentation outputs showed strong visual similarity to the ground truth masks.
- The model demonstrated stable convergence behavior during optimization.

The segmentation results indicate that the model successfully learned tumor-related spatial features from multi-modal MRI scans[6], [7].

The prediction outputs also demonstrate that the framework can preserve major tumor structures while reducing segmentation inconsistencies.

The combination of quantitative metrics and qualitative visualization provides a more comprehensive understanding of segmentation performance.

The obtained results confirm that the proposed framework is capable of generating meaningful tumor masks suitable for subsequent surgical planning stages including risk analysis and trajectory generation[1], [5].

5.7 System-Level Evaluation

Beyond segmentation evaluation, the proposed framework successfully integrated multiple processing stages within a unified pipeline.

The system combines:

- MRI preprocessing
- Tumor segmentation
- Functional mapping
- Risk-aware planning[1], [5]
- Interactive visualization

The integration of these stages allows the framework to support AI-assisted surgical planning workflows within a single processing environment.

The generated tumor masks can be integrated with atlas-based functional mapping and risk analysis components to support safer trajectory generation[5].

The framework architecture also supports interactive visualization and user interaction through the frontend application and backend processing pipeline.

The modular structure of the system improves maintainability and allows future expansion of additional processing components.

5.8 Limitations

Although the proposed framework demonstrated effective segmentation performance, several limitations remain.

One limitation involves dependence on dataset quality and annotation consistency. Variations in MRI quality or label accuracy may affect segmentation performance.

Another limitation relates to computational complexity because medical image segmentation and three-dimensional processing operations require significant computational resources.

In addition, further validation using larger datasets and additional testing scenarios may improve evaluation reliability.

Despite these limitations, the proposed framework demonstrated stable performance and meaningful segmentation capability within the conducted experiments.

5.9 Chapter Summary

This chapter presented the experimental results and evaluation of the proposed AI-assisted brain tumor surgical planning framework.

The chapter discussed dataset visualization, training performance, Dice score evaluation, qualitative segmentation results, and overall system-level observations.

The experimental results demonstrated stable optimization behavior, progressive improvement in segmentation performance, and successful tumor localization capability using the SegResNet-based framework.

The obtained segmentation outputs provide an important foundation for subsequent risk-aware surgical planning and trajectory optimization stages

Chapter 6
Conclusion and Future Work

6.1 Conclusion

This project presented an AI-assisted framework for 3D brain tumor surgical planning with risk-aware trajectory optimization. The proposed system integrated multiple medical imaging and computational stages into a unified workflow designed to support preoperative surgical planning and tumor analysis.

The framework utilized multi-modal MRI scans from the BraTS 2020 dataset and applied several preprocessing operations including reorientation, resampling, and intensity normalization to standardize MRI volumes before analysis.

The tumor segmentation stage was implemented using the SegResNet architecture, which generated tumor masks from preprocessed MRI scans. The obtained segmentation outputs demonstrated meaningful overlap with ground truth annotations and showed stable optimization behavior during training.

The evaluation process showed progressive improvement in validation Dice score throughout training epochs, while the training loss decreased consistently, indicating successful model learning and stable convergence behavior.

Beyond segmentation, the framework integrated atlas registration and functional cortex mapping to identify anatomically sensitive brain regions including motor, language, and visual cortical areas in addition to white matter tracts.

The system additionally generated voxel-wise surgical risk maps used during trajectory generation and path evaluation. Candidate surgical paths were generated and ranked according to several criteria including cumulative risk, path length, smoothness, and proximity to eloquent brain regions.

The project also integrated a Flutter-based mobile application with Firebase authentication and FastAPI backend communication to provide organized frontend interaction and modular backend processing.

The overall system demonstrated successful integration between MRI preprocessing, tumor segmentation, functional mapping, risk-aware trajectory planning, and interactive visualization within a unified environment.

The obtained results indicate that combining deep learning-based segmentation with anatomical risk analysis can support safer and more intelligent surgical planning workflows.

6.2 Future Work

Although the proposed framework demonstrated promising performance, several future improvements may further enhance the system capabilities and overall efficiency.

One possible improvement involves training the segmentation framework on larger and more diverse MRI datasets to improve model generalization and robustness across different clinical cases.

Future work may also focus on improving segmentation accuracy and boundary refinement to generate more precise tumor representations.

Additional improvements may include:

- Real-time processing optimization

- More advanced trajectory planning algorithms
- Enhanced risk analysis techniques
- Improved visualization quality
- Additional atlas integration
- Expanded mobile application functionality
- More interactive frontend features
- Optimization of backend processing performance

Future development may also include expanding the framework to support additional medical imaging applications beyond brain tumor surgical planning.

Further evaluation using additional experimental scenarios and larger datasets may additionally improve system reliability and validation quality.

The integration of more advanced AI models and improved visualization techniques may further enhance the ability of the framework to support intelligent neurosurgical planning systems in future research and development work.

References

[1] A. Sharma, R. Patel, and M. Gupta, "AI-Assisted 3D Brain Tumor Surgical Planning with Risk-Aware Trajectory Optimization," 2024.

[2] J. Smith, L. Wang, and H. Kim, "Brain Tumor Segmentation Using Deep Learning Techniques," *arXiv preprint* arXiv:2305.00257, 2023.

[3] M. Brown, S. Ahmed, and T. Lee, "Artificial Intelligence in Medical Imaging," *arXiv preprint* arXiv:2401.00587, 2024.

[4] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "No New-Net," *arXiv preprint* arXiv:1802.10508, 2018.

[5] D. Miller and K. Johnson, "Risk-Aware Trajectory Optimization for Surgical Planning," *arXiv preprint* arXiv:2511.02893, 2025.

[6] R. Thomas, A. Verma, and P. Singh, "Medical Image Segmentation Techniques Using Deep Learning," *arXiv preprint* arXiv:2201.03777, 2022.

[7] L. Chen, Y. Zhao, and M. Ibrahim, "Atlas Registration and Functional Brain Mapping," *arXiv preprint* arXiv:2308.00128, 2023.

[8] S. Kumar and T. Wilson, "MRI Preprocessing and Normalization Methods," *arXiv preprint* arXiv:2104.00980, 2021.

[9] H. Garcia, P. Roy, and E. Adams, "Evaluation Metrics for Medical Image Segmentation," *arXiv preprint* arXiv:2011.00848, 2020.

[10] N. Taylor and J. Scott, "Interactive Visualization Techniques for Medical Imaging," *arXiv preprint* arXiv:2011.01045, 2020.

التخطيط الجراحي ثلاثي الأبعاد لأورام الدماغ بمساعدة الذكاء الاصطناعي مع تحسين المسار مع مراعاة المخاطر

تتطلب جراحات أورام المخ تخطيطًا دقيقًا للغاية لضمان إزالة أكبر قدر ممكن من الورم مع الحفاظ على الوظائف الحيوية للمخ. وتعتمد الطرق التقليدية للتخطيط الجراحي بشكل كبير على التفسير اليدوي لصور الرنين المغناطيسي، مما يجعل العملية مستهلكة للوقت وتعتمد بصورة كبيرة على خبرة الطبيب. ومع التطور السريع في تقنيات الذكاء الاصطناعي وتحليل الصور الطبية، أصبح من الممكن تطوير أنظمة ذكية قادرة على دعم الأطباء في التشخيص واتخاذ القرارات الجراحية.

يقدم هذا المشروع إطار عمل يعتمد على الذكاء الاصطناعي للمساعدة في التخطيط ثلاثي الأبعاد لجراحات أورام المخ مع تحسين مسارات الوصول الجراحي وفقًا لمستوى الخطورة. يجمع النظام المقترح بين تقسيم الأورام باستخدام التعلم العميق، وتحديد المناطق الوظيفية بالمخ، وتحليل المخاطر، بالإضافة إلى العرض ثلاثي الأبعاد داخل خط معالجة متكامل.

يعتمد النظام على صور الرنين المغناطيسي من مجموعة بيانات BraTS 2020والتي تشمل أنواع T1 و T1ceو T2وFLAIR، حيث يتم إجراء عمليات المعالجة المسبقة ثم استخدام نموذج لتقسيم الورم وتحديد أجزائه المختلفة بدقة.

ولزيادة مستوى الأمان الجراحي، يتم تحديد المناطق الوظيفية الحساسة بالمخ مثل مناطق الحركة واللغة والإبصار باستخدام خرائط تشريحية للمخ، ثم دمج هذه المناطق داخل خريطة مخاطر ثلاثية الأبعاد تمنح قيم خطورة مختلفة لكل جزء من أجزاء المخ. بعد ذلك يتم إنشاء عدة مسارات جراحية وتقييمها اعتمادًا على طول المسار ومستوى الخطورة وزاوية الدخول باستخدام خوارزمية تحسين تعتمد على الوعي بالمخاطر.

كما يوفر النظام أدوات تفاعلية للعرض ثلاثي الأبعاد تسمح للطبيب باستكشاف الورم والمناطق الوظيفية والمسارات المقترحة داخل واجهة ويب متكاملة. ويهدف هذا المشروع إلى دعم جراحي المخ من خلال تحسين كفاءة التخطيط الجراحي وتقليل احتمالية الإضرار بالمناطق الحيوية بالمخ وتحسين عملية اتخاذ القرار الطبي .

DELTA UNIVERSITY
FOR SCIENCE AND TECHNOLOGY
FACULTY OF ARTIFICIAL INTELLIGENCE

جامعة الدلتا للعلوم والتكنولوجيا
كلية الذكاء الإصطناعي
قسم الذكاء الإصطناعي

التخطيط الجراحي ثلاثي الأبعاد لأورام الدماغ بمساعدة الذكاء الاصطناعي مع تحسين المسار مع مراعاة المخاطر

مشروع مهني قائم على العمل في مجال الذكاء الاصطناعي - AI317 (2)

مقدم من:

توفيق محمد توفيق	4231005
أدهم أسامه الرفاعي	4232010
بسمله هاشم عبدالرحمن	4231371
أروى هشام عبدالعزيز	4231468
مريم صلاح الدين السيد	4231469
ملك عرفه حسين	4231503

تحت اشراف:
ا. د. هناء سالم محمد سالم
نائب عميد
كلية الذكاء الاصطناعي لشؤون التعليم وشؤون الطلاب

الفصل الدراسي الربيعي
2026\2025